

Time spent: 0 sec

Subject:
Physics

QID: 400297

System:
Mechanics and Energy

Osteoporosis is a degenerative bone condition in which healthy bone degenerates into porous bone. Because bone provides the structural integrity that allows the musculoskeletal system to bear weight, osteoporosis is associated with a higher risk of bone fracture.

Researchers conducted an experiment designed to assess the strength and elasticity of osteoporotic bone relative to normal bone. Osteoporotic and normal bone samples with identical physical dimensions were collected and placed within an pneumatic compression device, as shown in Figure 1.

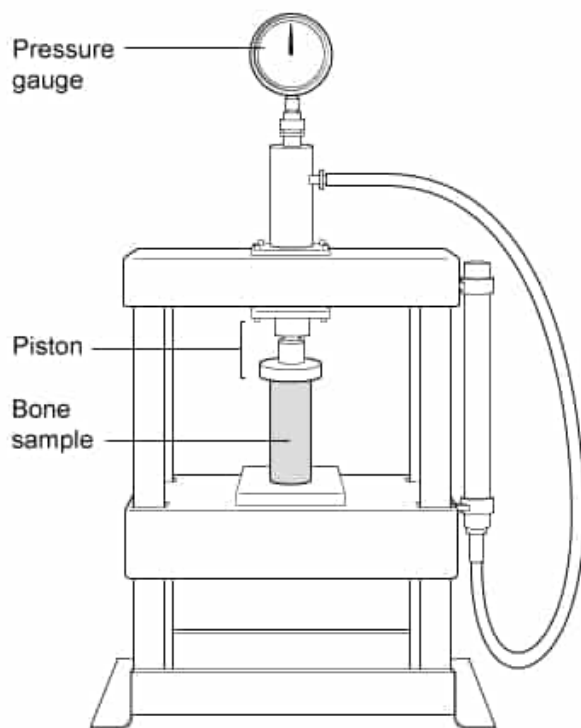


Figure 1 Bone compression device loaded with a bone sample

The force required to compress the bone samples longitudinally (lengthwise) by a given distance was measured, with the results shown in Figure 2.

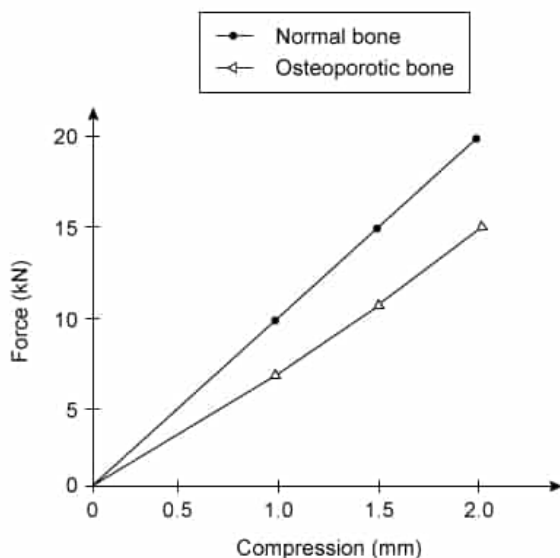


Figure 2 Relationship between force and compression for normal and osteoporotic bones

Researchers observed that bone samples returned to their original dimensions for deformations not exceeding 1% of the original length L_0 . For each deformation, the associated strain ε is defined as the change in length of an object ΔL divided by the object's original length (Equation 1).

$$\varepsilon = \frac{\Delta L}{L_0}$$

Equation 1

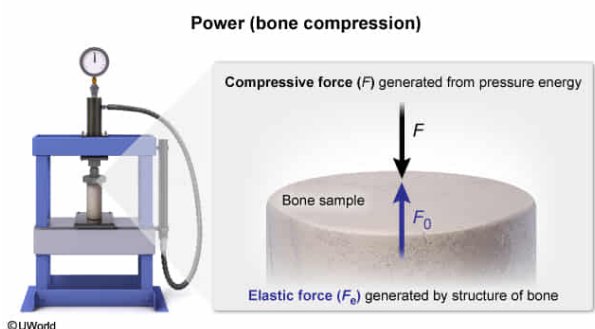
Based on these observations, researchers concluded that both osteoporotic bone and normal bone behaved as an elastic material under a wide range of applied forces.

If maintaining the maximal compression of an osteoporotic bone sample for 5 s requires 80 J of energy, what is the power expended by the compression machine?

- ☐ A. 8 W
- ✓ ☒ B. 16 W [90%]
- ☐ C. 80 W [2%]
- ☐ D. 400 W [6%]

Correct
90% answered correctly

Explanation:



Power (P) describes the **expenditure of energy** through physical processes that do not take place instantaneously. In mechanical systems, power is expressed in terms of the quantity of **work (W)** done on or by a system **over a period of time (t)**:

$$P = \frac{W}{t}$$

Because work is a product of **force (F)** acting over a distance (d), power expenditure within mechanical systems is usually discussed in terms of objects being moved through space. However, generating forces through **electrical power** (or other means) also expends energy. As such, power may also be expressed in terms of energy (E) per unit time:

$$P = \frac{E}{t}$$

In the passage, a compression device is used to compress bone samples and yield displacement away from an equilibrium point. Initially, the forces that compress the bone samples complete some mechanical work as the bone surface of the bone is displaced some distance from its equilibrium point.

Once the bone is fully compressed, no further mechanical work takes place, but the compression device must still expend energy to counteract the elastic force of the bone. If maintaining the compression of the bone for 5 s requires an energy expenditure of 80 J by the compression device, the power required during the time interval is:

$$P = \frac{80 \text{ J}}{5 \text{ s}} = 16 \frac{\text{J}}{\text{s}} = 16 \text{ Watts (W)}$$

Therefore, keeping an osteoporotic bone sample fully compressed for 5 s corresponds to a power expenditure of 16 W by the compression device (**Choice B**).

Educational objective:

Power describes the expenditure of energy over a period of time. Power may describe the expenditure of energy through a variety of energy types, including mechanical, electrical, and thermal.

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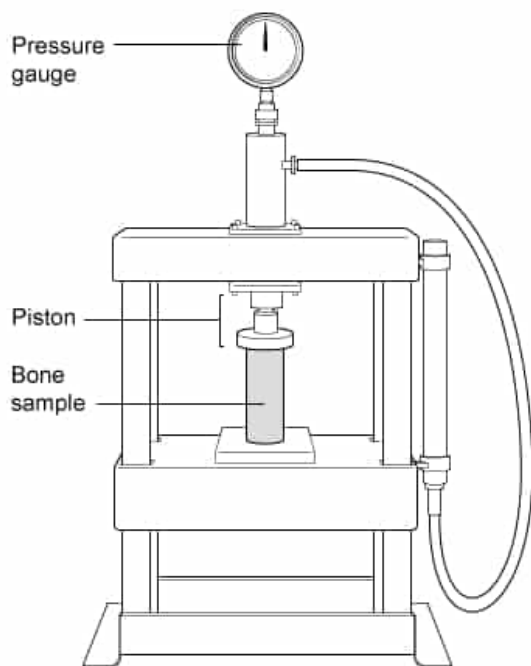


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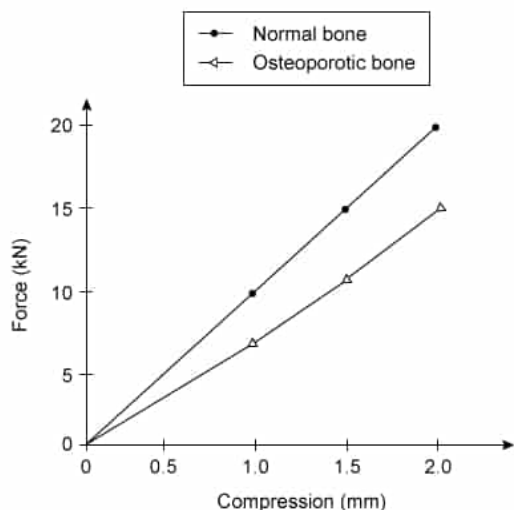


Figure 2 Relationship between force and compression for normal and osteoporotic bones

Researchers observed that bone samples returned to their original dimensions for deformations not exceeding 1% of the original length L_0 . For each deformation, the

associated strain ε is defined as the change in length of an object ΔL divided by the object's original length (Equation 1).

$$\varepsilon = \frac{\Delta L}{L_0}$$

Equation 1

Based on these observations, researchers concluded that both osteoporotic bone and normal bone behaved as an elastic material under a wide range of applied forces.

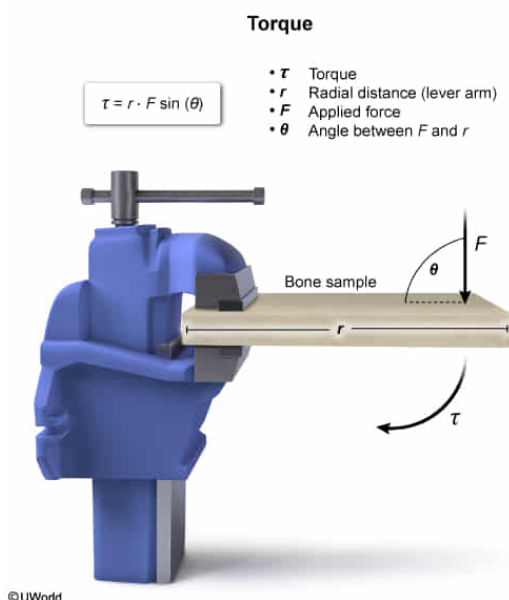
Researchers devised a different experiment to measure the torque required to fracture a long bone fixed in place at one end. What information is required to calculate the torque?

- ☐ A. Applied force, bone density, and angle of the applied force [2%]
- ✓ ☒ B. Applied force, bone length, and angle of the applied force [94%]
- ☐ C. Bone density, angle of the applied force, and bone length [2%]
- ☐ D. Bone density, applied force, and bone length [1%]

Correct

94% answered correctly

Explanation:



A **torque (τ)** is the application of force to an object around a **pivot point** (a fixed point such as a hinge or axle, often seen in mechanical systems) along a lever arm. Because torques promote the rotation of an object about the pivot point, torques are described as rotational forces.

The magnitude of a torque τ is expressed in terms of the magnitude of the **applied force (F)**, the radial distance (r) between the pivot point and the location at which the translational force is applied, and the **angle (θ)** between F and r .

$$\tau = r \cdot F \sin(\theta)$$

A **lever arm** separates the pivot point from the location of the applied force (eg, a wrench). Consequently, the lever arm is a physical distance that transmits the torque to the pivot point such that the length of the lever arm corresponds to radial distance r .

In this question, researchers fix a long bone at one end and apply a translational force at some distance away from the fixed end. In this setup, the lever arm is the length of bone that separates the pivot point at the fixed end from the point of the applied force.

Therefore, the length of the bone, the magnitude of the translational force applied, and the angle between the bone and the translational force are necessary to determine the magnitude of the torque generated during the experiment.

(Choices A and C) Knowledge of bone density would be essential only in discussing density-dependent properties such as the buoyancy of bone or the strength of bone relative to its weight.

(Choice D) The angle separating the translational force from the lever arm is essential for calculating the torque because any angle other than 90° yields a value less than 1, decreasing the torque transmitted to the pivot point.

Educational objective:

An applied force that promotes the rotation of an object around a pivot point is known as torque. The magnitude of the applied force, the length of the lever arm, and the angle between the applied force and the lever arm are all needed to calculate the torque.

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QID:400299

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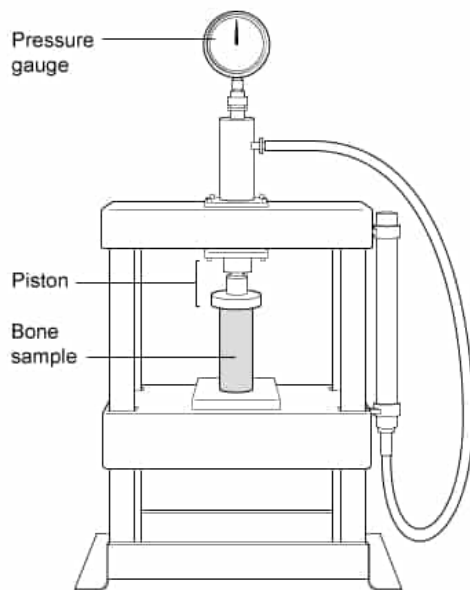


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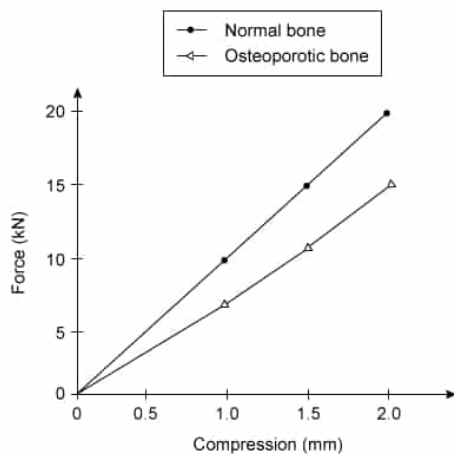


Figure 2 Relationship between force and compression for normal and osteoporotic bones

Researchers observed that bone samples returned to their original dimensions for deformations not exceeding 1% of the original length L_0 . For each deformation, the associated strain ϵ is defined as the change in length of an object ΔL divided by the object's original length (Equation 1).

$$\epsilon = \frac{\Delta L}{L_0}$$

Equation 1

Based on these observations, researchers concluded that both osteoporotic bone and normal bone

behaved as an elastic material under a wide range of applied forces.

Which of the following expressions is equal to U_{el} , the elastic potential energy stored within a compressed bone?

- ✓ ☒ A. $\frac{1}{2}k(\varepsilon \cdot L_0)^2$ [84%]
- ☐ B. $\frac{1}{2}k\left(\frac{\varepsilon}{L_0}\right)^2$ [10%]
- ☐ C. $2k(\varepsilon \cdot L_0)^2$ [3%]
- ☐ D. $2k\left(\frac{\varepsilon}{L_0}\right)^2$ [1%]

Correct

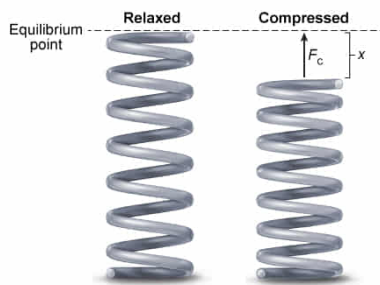
85% answered correctly

Explanation:

Elastic potential energy (spring)

$$U_{el} = \frac{1}{2}kx^2$$

- U_{el} Elastic potential energy
- k Spring constant
- x Displacement from equilibrium point
- F_c Force resisting compression



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Stretching or compressing an elastic object away from its **equilibrium point** (ie, its position at rest) increases its **elastic potential energy** (U_{el}), which is a form of energy that depends on the configuration of elastic objects. U_{el} is stored within an elastic object because **work** must be done against the elastic force F_{el} generated when an elastic object is displaced. Consequently, U_{el} quantifies the capacity of an elastic object to do work in the future using energy stored by work completed on the object in the past.

The magnitude of potential energy stored within elastic, spring-like objects is a function of the **displacement from equilibrium** (x) and the **spring constant** (k), a measure of the relative elasticity or stiffness of the spring-like object:

$$U_{el} = \frac{1}{2}kx^2$$

For the experiment described in this question, x is equal to the change in length (ΔL) of the bone sample because the initial length of the bone sample L_0 corresponds to the equilibrium position prior to compression. Making this substitution into Equation 1 and solving for x gives:

$$\varepsilon = \frac{\Delta L}{L_0} = \frac{x}{L_0}$$
$$x = \varepsilon \cdot L_0$$

Because the passage states that compressed bone acts like a spring-like object, the above expression for displacement can be substituted into the equation for elastic potential energy to yield:

$$U_{\text{el}} = \frac{1}{2}k(\epsilon \cdot L_0)^2$$

Therefore, the magnitude of the potential energy stored within a compressed bone sample may be expressed as a function of the strain (ϵ) experienced by that sample **(Choice A)**.

Educational objective:

Elastic potential energy describes the potential energy that results from displacing springs and other elastic objects away from their equilibrium point. Elastic potential energy is related to the spring constant and the magnitude of displacement from equilibrium point.

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